

UNLOG



TỐNG GIA LINH



LÊ VŨ HẠNH THẢO



LÊ THỊ LAN



VŨ HUY QUÂN

EXECUTIVE SUMMARY



SITUATION

The company's fulfillment network is operating under a highly concentrated demand structure while demand is geographically clustered around key urban and regional markets, and customer channels require sharply different fulfillment models. My Phuoc currently carries the bulk of volume, while Vinh Loc faces limitations as B2C and e-commerce demand scale.

COMPLICATION

A single warehouse and inventory policy risks creating channel conflict, inefficient picking, excess safety stock, and service failures

My Phuoc cannot meet fast urban SLA requirements without an urban dark node, Vinh Loc alone cannot support a true citywide 2-hour promise

Inventory must shift from separated channel stock to a controlled virtual-pooling model to reduce safety stock while protecting priority customers.

QUESTION

How should the company redesign its omni-channel distribution system — from SKU profiling, RDC network allocation, inventory pooling, to warehouse picking operations — to serve both B2B bulk orders and B2C/E-commerce fast delivery without increasing stockout risk or operational cost?

ANSWER

Build differentiated fulfillment operations for each customer segment

Redefine warehouse roles by channel and service promise

Optimize inventory through velocity-based slotting and virtual pooling

IMPACT

Reduce safety stock burden by 61.8%

Lift SLA to 98% level

THE NETWORK IS STRUCTURALLY IMBALANCED AT EVERY LEVEL.

ALL TASKS



SKU Concentration — ABC-Frequency Matrix

[Q1.1 — ABC-XYZ Analysis]

Heatmap: SKU count by volume × pick frequency

	A	B	C		
A	19	6	3	470 total SKUs	19 AA fast SKUs
B	20	18	1		
C	14	60	329	329 CC dead weight	34 BA/CA labor trap

Concentration evidence

- 19 SKUs (4.04%) drive 57.87% of outbound quantity — 163,013 units — and 25.88% of all order lines (5,381 orders).
- 329 SKUs (70%) are slow and infrequent; candidates for deep reserve storage or SKU rationalization.
- 34 BA/CA SKUs are picked often but low-volume, creating disproportionate labor cost.

Operational implication: assign 19 fast movers to ergonomic pick-faces near packing stations with dedicated replenishment lanes.

Demand Volatility — ABC-XYZ Volatility Matrix

[Q1.1 — Volatility Layer]

	X	Y	Z	
A	1	15	12	Volatility classes X = Stable ($CV \leq 0.50$) Y = Seasonal ($0.50-1.00$) Z = Erratic ($CV > 1.00$)
B	1	27	11	
C	3	13	387	

Volatility diagnosis

- Class A SKUs: only 1 AX stable high-volume SKU vs 15 AY + 12 AZ — most high-volume items are volatile.
- Class B is dominated by 27 BY SKUs: medium-volume demand with seasonal patterns.
- Majority of high-volume SKUs sit in Y/Z — volatility, not just volume rank, should drive planning rules.

Dual-matrix purpose

ABC-Frequency → slotting & replenishment triggers
 ABC-Volatility → safety stock & forecast model selection

Warehouse Imbalance — Geographic Demand Split

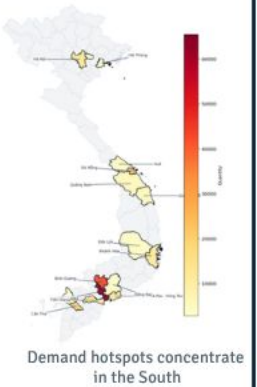
[Q1.2 — Distribution Heatmap]

Outbound scale: 281,704.73 units / 16,128.06 m³ / 21,400 rows

My Phuoc quantity share 93.02%

Vinh Loc quantity share 6.98%

Rows: **My Phuoc 75.43% vs Vinh Loc 24.57%**
 → Vinh Loc = high transaction frequency, low volume/order



Data & correction logic

- 97.12% row-level geographic coverage; 617 rows unresolved due to “unknown” ship-to customer in ERP.
- Resolved map shows My Phuoc at 92.59%; Statistical Scaling restores the true 93.02% vs 6.98% split.
- Regional concentration: Đông Nam Bộ 47.36%, Bắc Trung Bộ 25.49%, ĐBSCL 15.16%.

My Phuoc = bulk volume engine | Vinh Loc = transaction-complexity node

The warehouse imbalance is structural: My Phuoc handles bulk volume while Vinh Loc handles transaction complexity — these require different operational strategies, not a single unified model.

MODERN TRADE DEMANDS PALLET-LEVEL PRECISION ACROSS 34 PROVINCES. TRADITIONAL TRADE DEMANDS CARTON-LEVEL FLEXIBILITY ACROSS 60 PROVINCES. ONE WAREHOUSE POLICY CANNOT SERVE BOTH.



ALL TASKS

Geographic Demand Clusters - Top Provinces

[Q1.2 — Regional Demand Mapping]

Top 5 provinces by resolved quantity

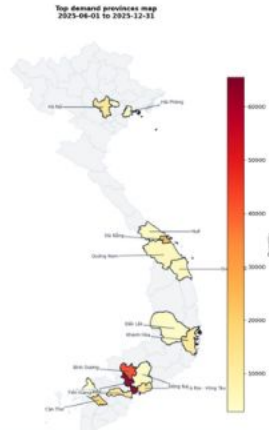
Hồ Chí Minh	1,402 orders / 65,527 units	27.47% qty
Bình Dương	661 / 43,012	18.03% qty
Đà Nẵng	296 / 24,340	10.20% qty
Cần Thơ	257 / 19,445	8.15% qty
Hà Nội	280 / 13,143	5.51% qty

Distance-demand signal

Close-range urban centers (HCMC, Bình Dương) order at **higher density** and **frequency** — distance inversely correlates with order size.

Overlapping regions

My Phuoc ships Fans and Vinh Loc ships Tefal to the same geographies — geography alone cannot define warehouse policy.

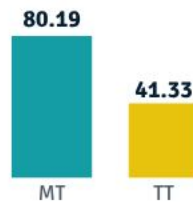


HCMC leads demand: 32.95% of resolved orders and 27.47% of resolved quantity.

Modern Trade vs Traditional Trade - Order Profile

[Q1.3 — Customer Segment Comparison]

Avg order qty



Avg order CBM



MT = large, consolidated, pallet-dominant | TT = small, fragmented, carton/loose-heavy

Dimension

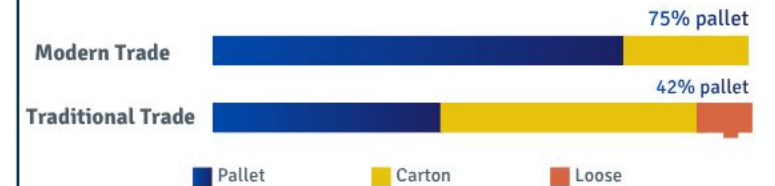
	MT	TT
Active customers	132	270
Avg order qty	80.19	41.33
Avg order CBM	5.16	2.13
SKU breadth/order	2.84	4.33
Province coverage	34	60
Avg distance	277.7	287.7
Pallet share	75.25%	41.70%
Carton share	22.52%	48.54%
Loose picking	2.23%	9.77%

Data story: MT has ~2× order size, while TT has wider geography and higher SKU breadth — different labor drivers.

Operational Implications by Segment

[Q1.3 — Fulfillment Design Implications]

Packaging unit mix



Modern Trade fulfillment

- Full-pallet pick → dedicated pallet staging lanes
- Higher LT sensitivity: 4,947.85 SS units/√ LT day
- Strict delivery slots → SLA contract risk if delayed
- Lower SKU breadth → simpler pick lists, higher volume/line

Traditional Trade fulfillment

- Mixed carton + loose picking → multi-zone routing
- Lower LT sensitivity: 3,962.68 SS units/√ LT day
- 60-province spread → fragmented outbound consolidation
- Higher SKU breadth → sorting and labeling complexity



WHY CHANGE? CURRENT RDC MODEL IS B2B-HEAVY AND STILL MISSES FAST B2C SLA

TASK 1 & TASK 2

Task 1 | Assessment and Flow reality

Existing two RDCs are not operating simultaneously



Strengths

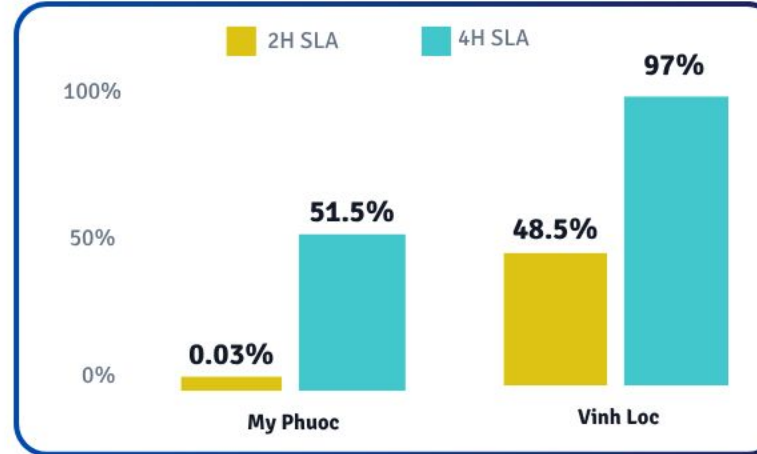
- Physical locations bracket the **Eastern-Western commercial corridor** of HCMC
- Both RDCs handle **mixed-channel flows** → operational flexibility

Weaknesses

- Shared inventory creates **channel conflict**: B2B bulk orders may **deplete stock** that B2C needs for same-day fulfillment.
- Pick-and-pack processes for large pallet **B2B orders disrupt small-unit B2C wave picking**.

Task 2 | SLA Baseline

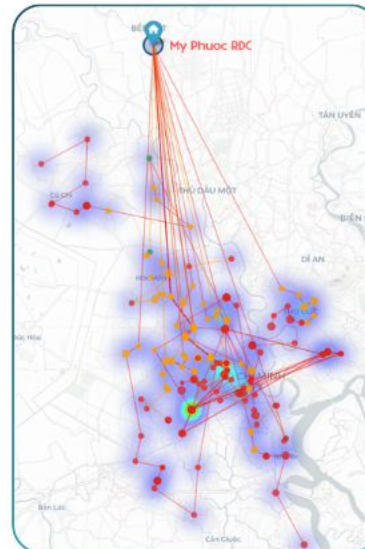
Demand-weighted SLA coverage under milk-runs



Vinh Loc is a stronger urban fulfillment base with higher SLA in both 4H and 2H criteria

Both warehouses cannot support 2H delivery requirement

My Phuoc is too distant to support B2C and Ecommerce demand without darknodes/hub



Prioritize Vinh Loc for urban B2C/E-commerce and My Phuoc for B2B/regional replenishment

My Phuoc strategically needs dark stores, while Vinh Loc maintains viable SLAs. However, long-term B2C and E-commerce growth → dark stores necessary for both



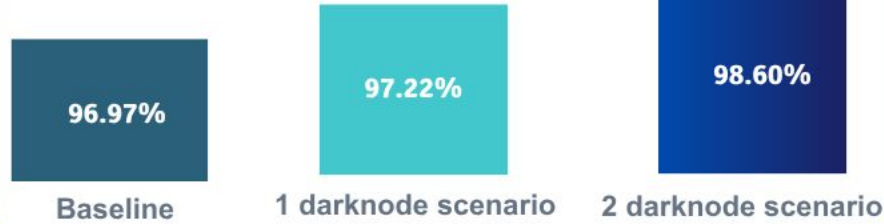
THE SOLUTION FOR FULLFILLMENT NETWORKS USING K-MEANS DEMAND CLUSTERING

TASK 3

My Phuoc RDC: 1 darknodes required



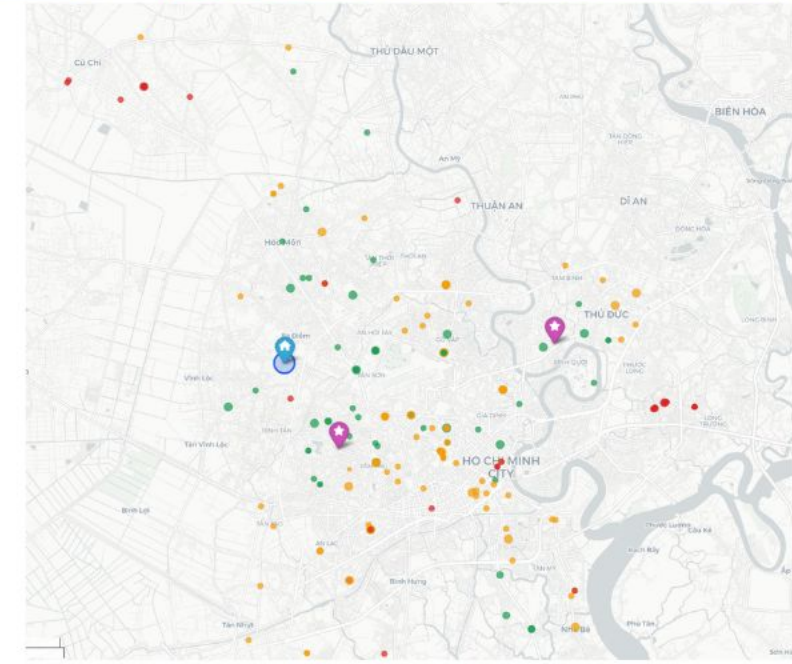
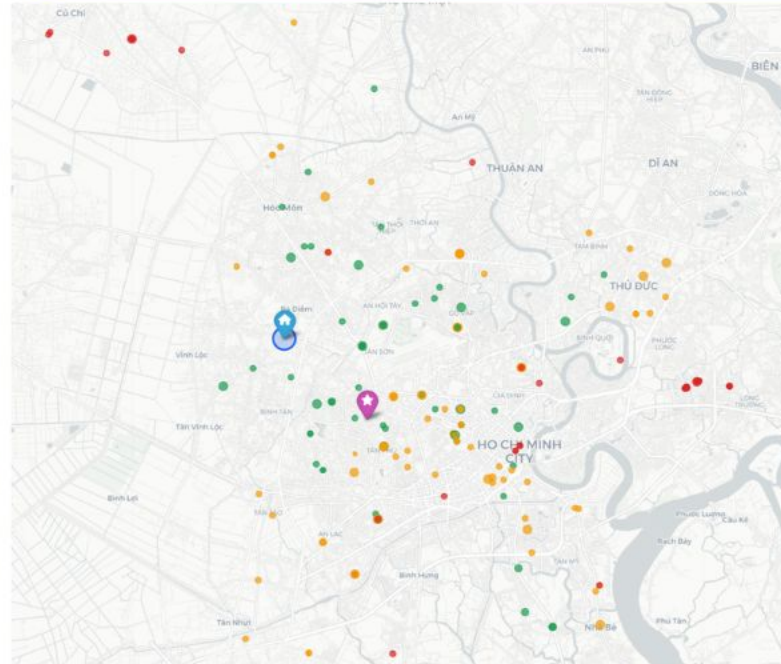
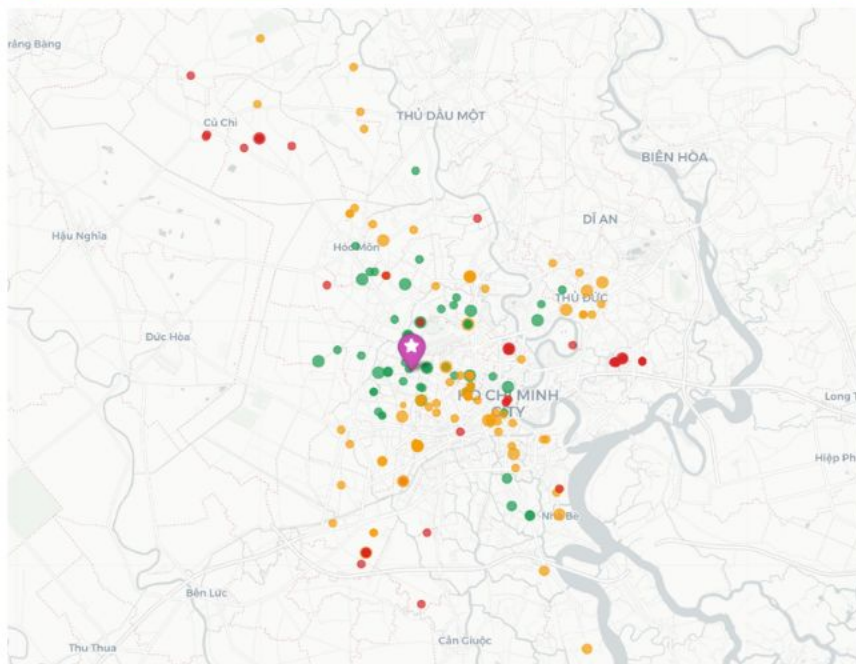
Vinh Loc RDC: 2 darknodes required



Scenario 1:

-(10.83627, 106.72860)
423 Kha Vạn Cân, Hiệp Bình
-(10.784616, 106.6204)
442B Nguyễn Sơn, Phú Thọ Hòa

Scenario 2: (10.7989, 106.63172)
33 Đường số 2, Phú Thọ Hòa



Node location

204 Tân Sơn Nhì, HCMC
10.79895755, 106.6317694

Phase 1: Improve immediately
My Phuoc RDC + 1 Darknode

Phase 2: Low capex, near 98% SLA
Vinh Loc RDC + 1 Darknode

Phase 3: Scale-up
Vinh Loc RDC + 2 Darknode



THE SOLUTION FOR INVENTORY OPTIMIZATION

TASK 4 & 5

1 | Lead-time reality: Demand is not a uniform lane

1.94 days

mile-weighted avg lead time (days)

49%

orders in Southeast near-RDC lane

4 days

days for northern / remote lanes

Insight: the average is pulled close to 2 days because almost half of orders sit in the 1-day Southeast lane, but the tail still requires protection

Region mix used for LTavg

Lane	Orders	LT	Share
Southeast	2,614	1d	49.0%
Central	1,027	3d	19.3%
Mekong Delta	800	2d	15.0%
North + remote	646	4d	12.1%
Highlands	247	2d	4.6%

2 | Inventory math: Safety stock grows with LT

$$SS = Z \times \sigma_{daily} \times \sqrt{LT_{avg}}$$

$$ROP = \mu_{daily} \times LT_{avg} + SS$$

$Z = 1.645$ for 95% service level

Result for 28 Class A SKUs: 12,168 units of required safety stock under the mile-weighted pool model.

3 | Operating model

1. MT contractual POs

2. MT E-commerce SLA

3. TT large distributors

4. TT small / walk-in

Control rules to prevent stock conflict

R1 – Common Pool: All SKU stock merged into a single WMS pool; OMS tracks virtual availability per channel.

R2 – Per-Channel ROP Floor: Each channel has its own ROP threshold; replenishment triggers when any channel hits its floor

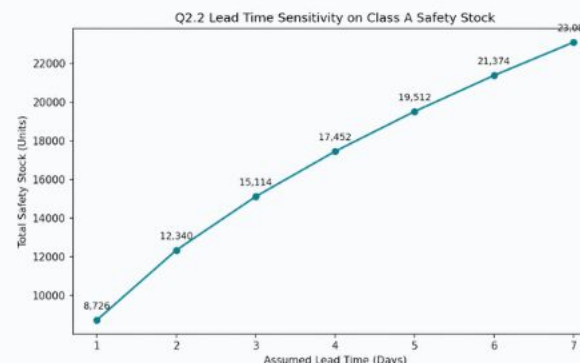
AMBER

Pool < Σ ROPs

RED

Pool < total SS

Action: prioritize confirmed MT POs, freeze spike-risk AZ SKUs for TT, and trigger emergency replenishment.



The lever is lead-time control: moving closer to 2 days keeps buffer inventory materially lower.

CONSULTING TAKEAWAY

Do not split inventory physically by B2B vs. B2C. Keep a virtual pooled stock in WMS/OMS, but govern it with channel-level ROP floors and escalation rules.

12,168
SS units

95%
service level

Daily
rebalance



3. WAREHOUSE OPERATIONS SOLUTIONS

3.1. SLOTTING OPTIMIZATION - TRAVEL TIME REDUCTION

Model Assumption

ZONE A — Pick-Face (Golden Zone)

28 SKUs | 5,767 picks | 8 m from packing | 92,272m round trip | 4 pcs/min (pick rate)

- Pick mode: Loose + carton break-bulk
- Slot rule: Highest order frequency → nearest slots; golden zone next to packing.

ZONE B — Forward Reserve

39 SKUs | 7,945 picks | 25 m from packing | 397,250 m round trip | 2.5 pcs/min pick rate

- Pick mode: Carton picking + replenish A
- Slot rule: Medium velocity → mid bay; min/max replenishment feeds the A pick-face.

ZONE C — Reserve / Bulk

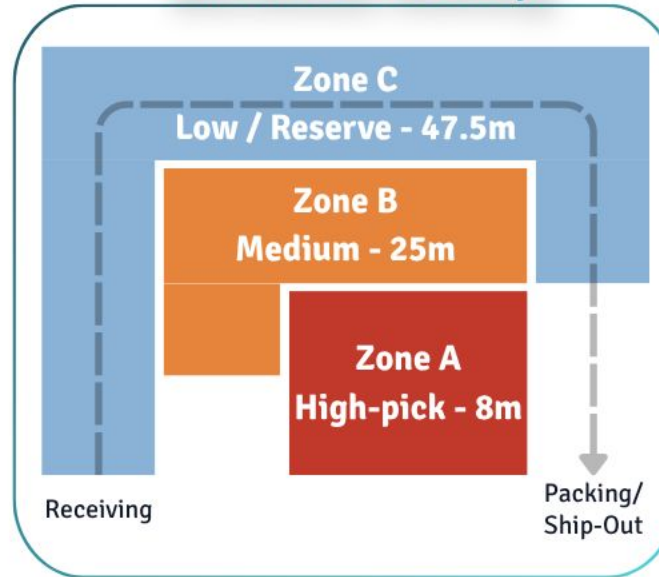
403 SKUs | 7,082 picks | 47.5 m from packing | 672,790 m | 2 pcs/min pick rate

- Pick mode: Pallet store, pick on demand
- Slot rule: Low velocity → far/upper rack, bulk block-stack.

35.4%

REDUCTION IN PICKER TRAVEL TIME ✓ EXCEEDS 30% TARGET

Warehouse Heatmap



Physical-distance model

Walk speed: 72 m/min (1.2 m/s, standard warehouse)

$$\text{Total round-trip distance} = \sum_{\text{zone}} \text{picks}_{\text{zone}} \times D_{\text{zone}} \times 2$$

$$\text{Travel time (min)} = \frac{\text{Total round-trip distance}}{\text{walk speed}}$$

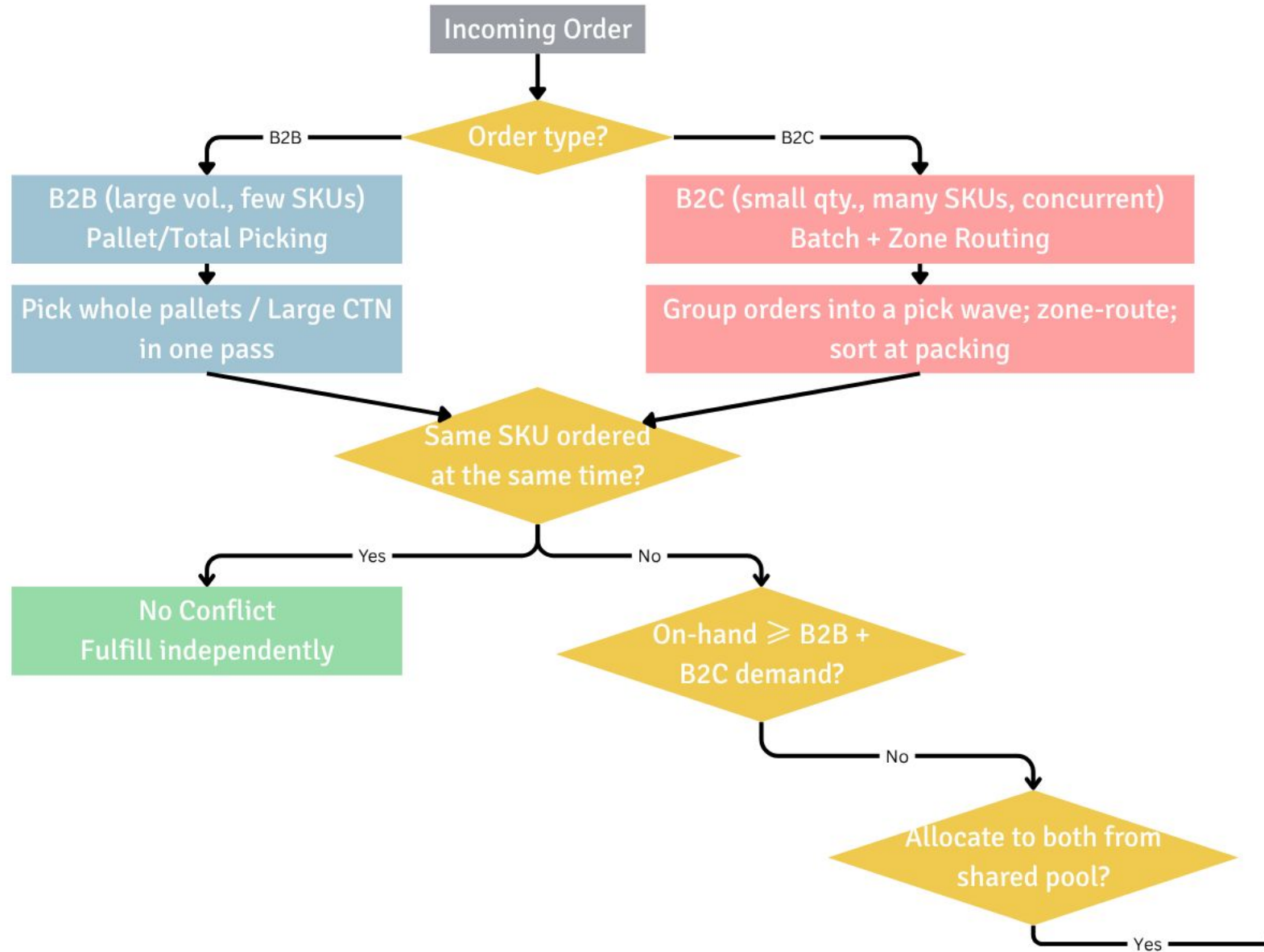
Baseline random replacement

$$\bar{D}_{\text{random}} = \frac{N_A \cdot D_A + N_B \cdot D_B + N_C \cdot D_C}{N}$$

Metric	Random storage	ABC Optimized
Avg distance / pick	43.3 m	27.9 m
Total round-trip dist	1,799,920 m	1,162,312 m
Total travel time	24,999 min	16,143 min



3.2. OMNI-CHANNEL PICK-AND-PACK PROCESS DESIGN



CONFLICT RESOLUTION

Same SKU ordered by B2B & B2C simultaneously

Stock sufficient?
→ Allocate both from shared pool

If stock insufficient:

- R1 Commit B2B PO first**
Contractual priority; penalty risk
- R2 Hold B2C safety buffer**
Prevents storefront stockout display
- R3 Split if B2C SLA critical**
Partial now; backorder remainder (2-4h SLA)
- R4 Emergency replenishment**
Trigger inter-warehouse transfer if below ROP